

Software Bid / Project Teams

UCS503 — Software Engineering Lab

Group	3C73	Date	13-08-2026
Team Name	SpatialX	Team ID	_____

Preferred Team Members

Name	Roll No.	Project Experience	Programming Languages
Arnav Mittal	1024031044	Data analysis automation; academic projects	Python, C++, SQL, Django
Pranjal Malhotra	1024031040	Web scraping; computer vision	Python, C++, SQL, Java

Programming Language / Environment Experience

List the languages the team is most comfortable developing in, in order of preference.

1. Python
2. C++
3. Java

Project Choices

Preference	Project Name	Unique Selling Point
First Choice	RoomSpace - Constraint-Based Room Planning and Decoration Recommendation System	A software system for planning room layouts and decoration from a room photograph, physical dimensions, and user-defined requirements. It produces practical layout alternatives rather than only textual interior-design suggestions.
Second Choice	Government Bureaucracy Translator	A RAG-based platform that gathers official government information and turns complex procedures, forms, and requirements into clear, tailored instructions, including eligibility, required documents, next steps, and official sources.
Third Choice	Context Checker	An NLP-based evidence-analysis tool that verifies claims by collecting and weighing supporting and contradicting evidence, then presents a transparent confidence level and source trail instead of a simple fake/real verdict.
Fourth Choice	Emergency Guidance System	A decision-support system for unexpected situations such as lost passports, stolen wallets, cancelled flights, or missing documents. It provides a structured action plan with relevant authorities, immediate steps, required documents, and official sources.

Additional Remarks / Inputs

Why RoomSpace? We would like to implement RoomSpace because it brings together **software engineering, computer vision, spatial reasoning, and an interactive user experience** in one practical system. Our aim is to build more than a recommendation application: the system should understand room constraints, generate feasible layouts, evaluate them, and let users iteratively refine their designs.

As our second choice, the Government Bureaucracy Translator interests us because it applies retrieval-augmented generation to a genuinely useful problem. It offers hands-on experience with document processing, retrieval, API design, and source verification while keeping the initial scope manageable.